Literature Review and Evaluation

MediCare - Evidence, Lecturer Feedback, Improvements, and Proposed Grading

DEPI Full Stack .NET Track | Submission deadline: 16 October 2026

Literature Review

Review focus

This focused review considers evidence on web-based medical appointment scheduling, missed appointments, reminders, and digital-health implementation. It is a project-oriented summary, not a systematic review of all available research.

Evidence summary

Online appointment access. Zhao et al. reviewed 36 articles describing 21 web-based medical appointment systems. Several studies reported improvements such as lower no-show rates, reduced staff work or waiting time, and improved satisfaction. The review also identified cost, flexibility, safety, system integrity, and patient adoption as implementation barriers. MediCare should support patient self-service while keeping the booking workflow accessible and reliable [1].

Missed appointments. Dantas et al. analyzed 105 studies selected from 727 retrieved articles. Their review found substantial variation across settings; longer lead time and previous missed appointments were frequently associated with non-attendance. MediCare should make dates and status visible and provide policy-compliant cancellation/rescheduling paths. The findings do not establish a universal no-show rate for this project [2].

Reminders. A 2023 systematic review included 61 studies and found evidence supporting appointment reminders by SMS, phone, or mail, while effectiveness varied by setting and delivery method. Reminder functionality is reasonable to include, but its local effect should be measured rather than assumed [3].

Digital-health implementation. WHO’s Global Strategy on Digital Health emphasizes coordinated organizational, human, financial, and technological resources. MediCare planning should therefore cover roles, workflows, support, privacy, and implementation resources as well as software features [4].

Implications for MediCare

• Design booking around the patient’s choice of doctor and available time, with clear status and confirmation.

• Prevent double booking through server-side and persistence-layer checks, not only calendar UI behavior.

• Make cancellation and rescheduling understandable within confirmed clinic rules.

• Treat reminders as configurable and measurable; do not claim a specific reduction in missed appointments without local evaluation.

• Protect records through role and ownership checks, and use synthetic data in coursework demonstrations.

• Use simple onboarding and accessible screens because patient familiarity and communication preferences influence adoption.

Review limitations

The cited studies involve different countries, specialties, user groups, and implementation designs. Their outcomes cannot be transferred directly to a DEPI prototype or a specific clinic. Use them as design guidance and evaluate the project with its own test participants and synthetic data.

References

[1] Zhao P, Yoo I, Lavoie J, Lavoie BJ, Simoes E. (2017). Web-Based Medical Appointment Systems: A Systematic Review. Journal of Medical Internet Research, 19(4), e134. DOI: 10.2196/jmir.6747. https://www.jmir.org/2017/4/e134

[2] Dantas LF, Fleck JL, Cyrino Oliveira FL, Hamacher S. (2018). No-shows in appointment scheduling - a systematic literature review. Health Policy, 122(4), 412-421. DOI: 10.1016/j.healthpol.2018.02.002. https://pubmed.ncbi.nlm.nih.gov/29482948/

[3] Behavioural economic interventions to reduce health care appointment non-attendance: a systematic review and meta-analysis. BMC Health Services Research (2023). https://pmc.ncbi.nlm.nih.gov/articles/PMC10594857/

[4] World Health Organization. (2025). Global strategy on digital health 2020-2027. https://www.who.int/publications/i/item/9789240116870

Feedback and Evaluation

Lecturer assessment

No lecturer comments or assessment were supplied. This section is prepared for the lecturer to complete after reviewing the project. Proposed improvements below are team recommendations, not attributed to the lecturer.

Assessment item	Lecturer feedback / evidence
Strengths observed	To be completed by lecturer
Required corrections	To be completed by lecturer
Implementation assessment	To be completed by lecturer
Documentation and presentation comments	To be completed by lecturer
Assessor and date	To be completed by lecturer

Suggested Improvements

• Add automated tests for authorization and appointment conflict handling.

• Improve keyboard navigation, field labels, color contrast, mobile layout, and error messages.

• Add an audit history for sensitive record access and administrator actions if time permits.

• Clarify appointment statuses, time-zone assumptions, and cancellation rules.

• Add notification preferences and clear delivery-failure handling.

• Consider future health-data interoperability only after integration, privacy, and consent needs are specified.

• Run short usability sessions with representative users and use findings to refine workflows.

Final Grading Criteria

The official DEPI rubric was not supplied. This example weighting is only a planning proposal; replace it with the lecturer's official criteria when available.

Category	Proposed weight	Evidence
Documentation and analysis	25%	Proposal, literature review, requirements and design
Implementation	35%	Working features, architecture, data handling and role workflows
Testing and quality	20%	Test plan/results, defect handling, security and usability checks
Presentation and demonstration	20%	Clear explanation, demo, team contribution, answers to questions
Total	100%	Illustrative only; confirm against official rubric

4.4 Weekly & Milestone Evaluation

Instructor assessment tracking - project phases and four implementation weeks

This table is prepared for the lecturer to complete during each project review. Dates for the implementation weeks are to be confirmed by the team. No lecturer feedback or grade has been recorded.

Phase / Week	Planned deliverables	Status	Evaluation date	Instructor feedback
Phase 1 Planning & Requirements	Proposal, project plan, risk plan, KPIs, requirements	Planned (16 Oct 2026)	Pending	Pending Instructor Review
Phase 2 Analysis & Design	Architecture, ERD, UML diagrams, wireframes	Scheduled (06 Nov 2026)	Pending	Pending Instructor Review
Week 1 Setup & Users	N-tier setup, EF Core schema, identity, doctor search	Scheduled (Date TBD)	Pending	Pending Weekly Evaluation
Week 2 Booking	Booking calendar, conflict detection, doctor dashboard	Scheduled (Date TBD)	Pending	Pending Weekly Evaluation
Week 3 Medical Records	Medical history, digital prescriptions, notifications	Scheduled (Date TBD)	Pending	Pending Weekly Evaluation
Week 4 Admin & Release	Admin dashboard, analytics, deployment	Scheduled (Date TBD)	Pending	Pending Weekly Evaluation

Review notes

Actual status, review date, lecturer comments, and required corrections should be updated after each meeting.
Phase dates are taken from the previously supplied evaluation schedule; implementation week dates remain unconfirmed.

Instructor name: _____ Signature: _____